higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

FLUID MECHANICS N6

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This marking guideline consists of 7 pages.

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QUESTION 1

1.1 In turbulent flow particles move in a disorderly manner occupying different relative positions in successive cross section. (2)

1.2 The area of a structure divided by its perimeter

OR

The ratio of area of a structure and its perimeter (1)

1.3 1.3.1
$$Q_S = V_S \times A_S$$
$$= \frac{\pi(0{,}255)^2}{4} \times 3{,}6\checkmark$$
$$= 0{,}184\ m^3/s\checkmark$$ (2)

1.3.2
$$Q_P = Q_S + Q_R$$
$$Q_P = \frac{\pi(0{,}405)^2}{4} \times 1{,}8\checkmark$$
$$= 0{,}232\ m^3/s\checkmark$$
$$Q_P = Q_S + Q_R$$
$$0{,}232 = 0{,}184 + Q_R\checkmark$$
$$Q_R = 0{,}048\ m^3/s\checkmark$$ (4)

1.3.3
$$V_R = \frac{Q}{a}$$
$$= \frac{0{,}048 \times 4}{\pi \times 0{,}105^2}\checkmark$$
$$= 5{,}547\ m/s\checkmark$$ (2)

1.3.4
$$V_Q = \frac{Q}{a}$$
$$= \frac{0{,}232 \times 4}{\pi \times 0{,}285^2}\checkmark$$
$$= 3{,}635\ m/s\checkmark$$ (2)

1.4
$$V_1 = \frac{0{,}0458 \times 4}{\pi \times (0{,}125)^2} = 3{,}735\ m/s\checkmark$$
$$V_2 = \frac{0{,}0458 \times 4}{\pi \times (0{,}08)^2} = 9{,}118\ m/s\checkmark$$
$$\frac{Pr_1}{\rho g} + \frac{v^2}{2g} + z_1 = \frac{Pr_2}{\rho g} + \frac{v^2}{2g} + z_2\checkmark$$
$$\frac{160}{9{,}81} + \frac{3{,}735^2}{2 \times 9{,}81} + 0\checkmark = \frac{P_2}{9{,}81} + \frac{9{,}118^2}{2 \times 9{,}81} + 1{,}1\checkmark$$
$$Pr_2 = 114{,}612\ kPa\checkmark$$ (8)

[21]

QUESTION 2

2.1

$A = L \times B$
$= 4{,}5 \times 2{,}5$✓
$= 11{,}25\ m^2$✓

$P = 2(2{,}5) + 4{,}5$✓
$= 9{,}5\ m$✓

$m = \frac{A}{P}$
$= \frac{11{,}25}{9{,}5}$ ✓
$= 1{,}184\ m$✓

$i = \frac{1}{2\ 550}$
$= 0{,}392 \times 10^{-3}$✓

$Q = AC\sqrt{mi}$
$= 11{,}25 \times 60 \times \sqrt{1{,}184 \times 0{,}392 \times 10^{-3}}$ ✓
$= 14{,}546\ m^3/s$✓ (9)

2.2

$V_A = \sqrt{\frac{g x^2}{2y}}$
$= \sqrt{\frac{9{,}81 \times 1{,}4^2}{2(2{,}1 - h)}}$ ✓
$= \sqrt{\frac{9{,}6138}{2{,}1 - h}}$ ✓

$V_{th} = \sqrt{2gh}$
$= \sqrt{2 \times 9{,}81 \times h}$✓
$= \sqrt{19{,}62h}$

$C_V = \frac{V_A}{V_{TH}}$✓

$0{,}9 = \frac{\sqrt{\frac{9{,}6138}{2{,}1 - h}}}{\sqrt{19{,}62h}}$ ✓

$0{,}81(41{,}202h - 19{,}62h^2) = 9{,}6138$

$h^2 - 2{,}1h + 0{,}606 = 0$✓

$h = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
$\frac{-(-2{,}1) \pm \sqrt{(-2{,}1)^2 - 4(1)(0{,}605)}}{2(1)}$ ✓
$h = 1{,}755$✓ or $0{,}345$ m✓ (10)

[19]

QUESTION 3

3.1

$$Q = ALSE\frac{N}{60}$$

$$= \frac{\pi(0{,}35)^2}{4} \times 0{,}55 \times 1 \times 1 \times 1{,}7\checkmark$$

$$= 0{,}08996\ m^3/s\checkmark$$

Without an air vessel:

$$Hf_d = \frac{4fl}{2gd}\left[\left(\frac{D^2}{d^2}\right)\ \omega\quad R\right]^2$$

$$= \frac{4 \times 0{,}0067 \times 40}{2 \times 9{,}81 \times 0{,}16}\left[\left(\frac{0{,}35^2}{0{,}16^2}\right) \times 1{,}7 \times \frac{0{,}55}{2}\right]^2\checkmark$$

$$= 1{,}709\ m\checkmark$$

$$P = \rho g Q\, h_f$$

$$= 1\ 000 \times 9{,}81 \times \frac{2}{3} \times 0{,}08996 \times 1{,}709\checkmark$$

$$= 1\ 005{,}449\ W\checkmark$$

With an air vessel:

$$Hf_d = \frac{4fl}{2gd}\left[\left(\frac{D^2}{d^2}\right)\ \frac{\omega}{\pi}\ R\right]^2$$

$$= \frac{4 \times 0{,}0067 \times 40}{2 \times 9{,}81 \times 0{,}16}\left[\left(\frac{0{,}35^2}{0{,}16^2}\right) \times \frac{1{,}7}{\pi} \times \frac{0{,}55}{2}\right]^2\checkmark$$

$$= 0{,}0362\ m\checkmark$$

$$P = \rho g Q\, h_f$$

$$= 1\ 000 \times 9{,}81 \times 0{,}08996 \times 0{,}0362\checkmark$$

$$= 31{,}947\ W\checkmark$$

Power saved $= 1\ 005{,}449 - 31{,}947\checkmark$

$= 973{,}502\ W\checkmark$ (12)

3.2 3.2.1

$$Q_{th} = \text{ALSE}\frac{N}{60}$$

$$= \frac{\pi(0,5)^2}{4} \times 0,525 \times 2 \times 1 \times \frac{65}{60}\checkmark$$

$$= 0,223 \text{ m}^3/\text{s}\checkmark$$

$$\% \text{ slip} = \frac{Q_{th} - Q_A}{Q_{th}} \times 100\%$$

$$= \frac{0,223 - 0,0184}{0,223} \times 100\%\checkmark$$

$$= 91,762\%\checkmark$$

Alternative:

$$C_D = \frac{Q_A}{Q_{th}}$$

$$= \frac{0,0184}{0,223}$$

$$= 0,0824\checkmark$$

$$\% \text{ slip} = (1 - C_D) \times 100\%$$

$$= (1 - 0,0824) \times 100\%$$

$$= 91,761\%\checkmark$$ (4)

3.2.2

$$C_D = \frac{Q_A}{Q_{th}}$$

$$= \frac{0,0184}{0,223}\checkmark$$

$$= 0,0824\checkmark$$ (2)

3.2.3

$$\text{P} = \rho g Q_{th} h_T$$

$$= 10^3 \times 9,81 \times 0,223 \times 14\checkmark$$

$$= 30,675 \text{ kW}\checkmark$$ (2)

3.3 A – Suction stroke
B – Delivery stroke
C – Delivery acceleration head or H_{ad}
D – Delivery head or h_d
E – Suction acceleration head or H_{as}
F – Suction head or h_s
G – Atmospheric head or H_{at} (9)

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H – Suction friction head or h_{fs}
I – Delivery friction head or h_{fd}

(9 × 1)

3.4 3.4.1 $H_{pre} = H_{at} - h_s - h_{as}$

3.4.2 $H_{pre} = H_{at} + h_d - h_{fd}$

3.4.3 $H_{pre} = H_{at} - h_s + h_{as}$

(3 × 1) (3)

3.5
$$\left(\frac{N_2}{N_1}\right)^2 = \frac{wg_2}{wg_1}$$
$$\left(\frac{N_2}{500}\right)^2 = \frac{50}{45}\checkmark$$
$$N_2 = 527{,}046 \text{ r/min}\checkmark$$

$$\frac{Q_2}{Q_1} = \frac{N_2}{N_1}$$
$$Q_2 = \frac{N_2}{N_1} \times Q_1$$
$$= \frac{527{,}046}{500} \times 620\checkmark$$
$$= 653{,}537 \text{ m}^3/\text{s}\checkmark$$

This fan cannot be used since it will be supplying less than expected.✓
NOTE: NO mark if the answer is given without a reason. (5)
[37]

QUESTION 4

4.1
$$H = 0{,}93 \times 45\checkmark$$
$$= 41{,}85 \text{ m}\checkmark$$

$$V_A = C_V\sqrt{2gh}$$
$$= \sqrt{2 \times 9{,}81 \times 41{,}85}\checkmark$$
$$= 28{,}655 \text{ m/s}\checkmark$$

$$U_i = \frac{\pi D N}{60}$$
$$= \frac{\pi \times 3 \times 150}{60}\checkmark$$
$$= 23{,}562 \text{ m/s}\checkmark$$

$$\text{Sin}30° = \left(\frac{v_i}{28{,}655}\right)\checkmark$$
$$V_i = 14{,}328 \text{ m/s}\checkmark$$

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$$\beta_i = \tan^{-1}\left(\frac{14{,}328}{25 - 23{,}562}\right)\checkmark$$
$$= 84{,}269°\checkmark$$

$$E = \frac{U_I \times (V_{wI} - V_{wo})}{g}$$
$$= \frac{23{,}562 \times (20 - 5)}{9{,}81}\checkmark$$
$$= 36{,}028\ \text{m}\checkmark$$

$$\eta = \frac{E}{H}$$
$$= \frac{36{,}028}{45} \times 100\%\checkmark$$
$$= 80{,}061\%\checkmark \quad (14)$$

4.2 $$Q = \frac{43}{60} \text{ and } U = 17\ \text{m/s}$$
$$= 0{,}717\ \text{m}^3/\text{s}\checkmark$$

$$V_A = C_V\sqrt{2gh}$$
$$= \sqrt{2 \times 9{,}81 \times 60}\checkmark$$
$$= 34{,}31\ \text{m/s}\checkmark$$

$$P = \rho QU\ (V - U)\ (1 + \cos(180° - y))\checkmark$$
$$= 10^3 \times 0{,}717 \times 17 \times (34{,}31 - 17)\ [1 + \cos(180° - 162°)]\checkmark$$
$$= 411{,}473\ \text{kW}\checkmark$$

$$\eta = \frac{U}{gh}\ (V - U)\ (1 + \cos(180° - y)) \times 100\%\checkmark$$
$$= \frac{17}{9{,}81 \times 60}\ (34{,}31 - 17)\ [1 + \cos(180° - 162°)] \times 100\%\checkmark$$
$$= 97{,}545\%\checkmark \quad (9)$$

[23]

TOTAL: 100

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